

Continual Learning for Bangla NLP

Mitigating Catastrophic Forgetting in Cross-Lingual Transfer

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What is Bangla NLP?

Natural Language Processing (NLP): Teaching computers to understand human language

Example Tasks:

- Sentiment Analysis: Is this movie review positive or negative?
 - "এই সিনেমাটি দারুণ ছিল" (This movie was great) → Positive
- Named Entity Recognition: Find names of people, places, organizations
 - "ঢাকায় রহিম একটি কোম্পানিতে কাজ করে" → Rahim (Person), Dhaka (Location)

The Bangla Language:

- 7th most spoken language globally (230+ million speakers)
- Official language of Bangladesh and West Bengal, India
- Problem: Far less digital resources compared to English

The Resource Gap

Training an NLP Model: Like teaching a student

English (High-Resource):

- Millions of labeled examples
- “The food was delicious”
→ Positive (Sentiment)
- “Apple Inc. released iPhone”
→ Apple Inc. (Organization)
- Thousands more examples...

Bangla (Low-Resource):

- Tens of thousands of examples
- Much smaller variety
- Expensive to create more

Model struggles to learn

Model learns patterns very well

Key Question

Can we use knowledge from other languages to help Bangla?

Solution 1: Cross-Lingual Transfer Learning

Core Idea: Learn from a related language, then adapt to Bangla

Example: Hindi $\rightarrow$ Bangla Transfer

Both languages share similar grammar patterns:

- Bangla: আমি ভাত খেতে চাই (Ami bhaat khete chai)
- Hindi: मुझे चावल खाना है (Mujhe chawal khana hai)
- English: I want to eat rice

Pattern: Subject-Object-Verb order (similar in both)

The Process:

1. Train model on Hindi (lots of data available)
2. Fine-tune on Bangla (limited data)
3. Model already knows similar patterns $\rightarrow$ learns faster

But what if we want to use multiple languages?

The Problem: Catastrophic Forgetting

Definition: When learning something new, the model completely forgets what it learned before

Real Example Scenario:

1. Train on Hindi sentiment analysis
 - Test on Hindi $\rightarrow$ 90% accuracy
2. Now train on Tamil sentiment analysis
 - Test on Tamil $\rightarrow$ 85% accuracy
 - Test on Hindi $\rightarrow$ 45% accuracy (was 90%!)
3. Now train on Bangla
 - Test on Bangla $\rightarrow$ 80% accuracy
 - Test on Hindi $\rightarrow$ 30% accuracy
 - Test on Tamil $\rightarrow$ 35% accuracy

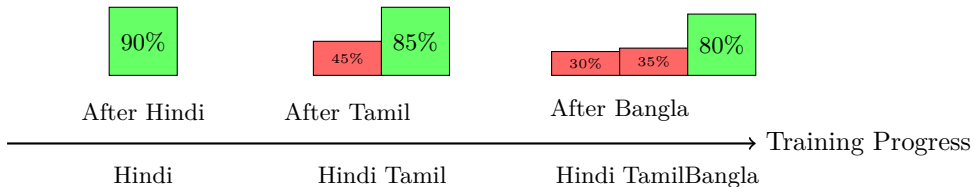
The Core Problem

Each new language overwrites previous knowledge instead of adding to it

Visualizing the Problem

Green = Good performance

Red = Forgotten (poor performance)



What We Want Instead: All three languages maintain high performance

Solution 2: Continual Learning

Core Idea: Learn new languages while remembering old ones

Simple Analogy:

- Catastrophic Forgetting = Studying for exams by completely forgetting previous subjects
- Continual Learning = Periodic review of old subjects while learning new ones

Basic Technique: Experience Replay

When training on a new language (e.g., Tamil):

- Mix 50% new Tamil examples
- Mix 50% old examples from Hindi (replay buffer)
- Model remembers Hindi while learning Tamil

Key Insight

Strategic mixing of old and new data prevents forgetting

Understanding Our Language Context

Why Study These Specific Indic Languages?

Indo-Aryan Family:

- Bangla (target language)
- Hindi: Most similar to Bangla
 - Same grammatical structure
 - Shared vocabulary roots
- Marathi: Moderately similar

Expected: Strong positive transfer

Dravidian Family:

- Tamil: Very different from Bangla
 - Different grammar rules
 - Different script
 - Different word order
- Telugu: Also Dravidian

Expected: Weaker transfer

Research Strategy:

- Test both similar and dissimilar languages
- Verify if linguistic similarity predicts transfer success
- Understand when continual learning helps most

Linguistic Similarity

Sentence: “I want to eat rice”

Language	Sentence	Word Order
Bangla	আমি ভাত খেতে চাই (Ami bhaat khete chai)	Subject-Object-Verb (I rice eat want)
Hindi	मुझे चावल खाना है (Mujhe chawal khana hai)	Subject-Object-Verb (I rice eat is)
Tamil	நான் சாதம் சாப்பிட விரும்புகிறேன் (Naan saadham saapida virumbugiren)	Subject-Object-Verb (Similar structure)
English	I want to eat rice	Subject-Verb-Object (Different!)

Key Observations:

- Bangla and Hindi: Nearly identical grammatical structure
- Tamil: Different family, but shares SOV word order
- English: Completely different word order (SVO)

Hypothesis: Hindi → Bangla transfer should work better than English → Bangla

Our Core Research Question

Central Problem

Which learning strategy produces the best Bangla NLP models when leveraging multiple source languages?

Three Competing Approaches:

1. Vanilla Transfer: Train on one language $\rightarrow$ Fine-tune on Bangla
 - Simple but ignores other useful languages
2. Cross-Lingual Multi-Source: Train on all languages together $\rightarrow$ Fine-tune on Bangla
 - Uses all languages but computationally expensive
3. Continual Learning: Sequentially train on languages $\rightarrow$ Apply forgetting mitigation
 - Efficient but requires preventing forgetting

Possible Thesis Problem

Continual learning with strategic language ordering can match multi-source performance while being more efficient and providing insights into linguistic transfer patterns

Hypothesized Performance Correlation

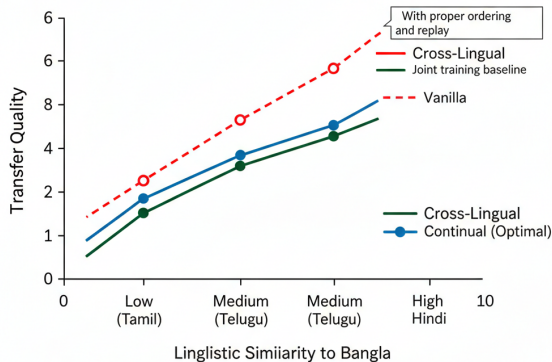


Figure: Transfer Learning correlation with linguistic similarity

Key Hypothesis: Continual learning captures language-specific patterns better than joint training,

RQ1: Quantifying Forgetting Patterns

- How much performance is lost on earlier languages?
- Does forgetting severity depend on language similarity?
- Are some NLP tasks more susceptible to forgetting?

RQ2: Optimal Language Curriculum

- Should similar languages (Hindi) come before or after distant ones (Tamil)?
- Does the optimal order differ across tasks?
- Can we design a general principle for language ordering?

RQ3: Linguistic Features as Predictors

- Do grammatical similarities (word order) predict transfer success?
- Does script similarity (Bangla vs Devanagari vs Tamil script) matter?

Assumptions and Scope

Core Assumptions:

1. Languages with similar grammar should transfer knowledge better
2. Experience replay can mitigate (but not eliminate) forgetting
3. Language ordering significantly impacts final performance
4. Linguistic similarity metrics can predict transfer quality

Research Scope:

- Included: 4 NLP tasks, 4-5 Indic languages, multiple orderings
- Excluded: Real-time learning, multilingual task transfer, novel architectures

Plan B: If Languages Show Unexpected Similarity

- If Tamil transfers better than expected → investigate linguistic features
- If Hindi transfers poorly → analyze structural differences
- Document unexpected patterns as novel findings

Three Research Streams:

1. Catastrophic Forgetting in Multilingual Continual Learning

- How do models forget across languages?
- What factors influence forgetting patterns?

2. Cross-Lingual Transfer Mechanisms

- How to leverage multiple source languages?
- What makes languages good transfer sources?

3. Continual Learning Techniques

- Experience replay and memory management
- Curriculum learning strategies

Key Paper 1: Catastrophic Forgetting in Multilingual CL

Winata et al. (2023) - Overcoming Catastrophic Forgetting in Massively Multilingual Continual Learning [4]

Key Findings:

- Studied forgetting across 51 languages on slot filling and intent detection
- Backward transfer: Performance drops on earlier languages (avg -16.90 F1)
- Forward transfer: New languages benefit from prior training (+4.45 F1)
- Language order does not significantly affect final performance

Proposed Solution: LR ADJUST

- Dynamic learning rate adjustment based on language position
- Reduces backward transfer from -16.90 to -3.59 F1 with A-GEM
- Works with replay-based methods (REPLAY, A-GEM)

Relevance: Direct evidence that forgetting is severe in multilingual settings

Key Paper 2: Experience Replay for Continual Learning

Rolnick et al. (2019) - Experience Replay for Continual Learning [3]

Core Concept: CLEAR (Continual Learning with Experience And Replay)

- Mix on-policy learning (new data) with off-policy learning (replay buffer)
- 50-50 split between new experiences and replayed experiences
- Behavioral cloning loss keeps policy close to historical data

Key Results:

- Almost eliminates catastrophic forgetting in RL tasks
- Works without task boundaries or task identity information
- Much simpler than prior methods (EWC, Progress & Compress)

Relevance: Provides strong baseline for replay-based continual learning

Key Paper 3: Continual Learning Under Language Shift

de Vries et al. (2025) - Continual Learning Under Language Shift [2]

Research Focus: How do transfer patterns emerge during sequential multilingual pre-training?

Main Findings:

- Forward transfer: Consistently positive and language-independent
- Backward transfer: Can be positive or negative, depends on language pair
- Increasing model size improves all languages but doesn't change transfer patterns
- Transfer effects correlate with language contamination and syntactic similarity

Key Metrics Introduced:

- Token Distribution Similarity (TDS): Cosine similarity of token frequency vectors
- Lang2vec distance: Pre-computed linguistic similarity

Relevance: Provides metrics to predict which languages will transfer well

Key Paper 4: Code-Switching Curriculum Learning

Yoo et al. (2024) - Code-Switching Curriculum Learning for Multilingual Transfer in LLMs [5]

Core Innovation: Mimic human language learning through curriculum

Three-Stage Training:

1. Token-level code-switching: Mix languages at word level
2. Sentence-level code-switching: Mix languages at sentence level
3. Monolingual corpora: Pure target language training

Results:

- Improves cross-lingual consistency (same answers across languages)
- Reduces spurious correlations between language and safety alignment
- Works for both high-resource (Japanese) and low-resource (Indonesian)

Relevance: Curriculum ordering matters for cross-lingual transfer

Key Paper 5: Multi-Source Cross-Lingual Transfer

Chen et al. (2019) - Multi-Source Cross-Lingual Model Transfer [1]

Central Question: How to leverage multiple source languages simultaneously?

Proposed Architecture: MAN-MoE

- Shared features: Learned via adversarial training (language-invariant)
- Private features: Language-specific experts (Mixture-of-Experts)
- Gating mechanism: Dynamically weights experts based on input similarity

Key Finding:

“Adversarial training alone is too restrictive—it only captures features common to all languages. When target language is similar to a subset of sources, language-specific features are crucial.”

Relevance: Multi-source transfer outperforms single-source for low-resource targets

What Literature Tells Us:

- Catastrophic forgetting is severe in multilingual settings
- Experience replay is effective
- Multi-source transfer works better than single-source

What We Don't Know (Our Contribution):

- Forgetting patterns specifically for Indic languages → Bangla
- Optimal language ordering for this language family
- How linguistic similarity (grammar, script) predicts transfer
- Task-specific forgetting dynamics in low-resource settings

Novel Contribution

First systematic study of continual learning for Indic language transfer to Bangla with linguistic feature analysis

Our Research Contributions:

1. Systematic Experimental Design

- Multiple language orderings
- Controlled baselines and ablation studies

2. Novel Analysis Framework

- Correlation between linguistic features and forgetting

3. Curriculum Optimization

- Evidence-based language ordering strategies
- Task-specific ordering recommendations

4. Benchmarking Contribution

- First continual learning benchmark for Indic languages
- Reproducible experimental protocols

Language Selection: Strategic Diversity

Target: Bangla (Indo-Aryan, Bangla script)

Source Languages (Intentional Diversity):

Language	Family	Script	Why Selected?
Hindi	Indo-Aryan	Devanagari	Closest grammatical relative to test best-case transfer
Marathi	Indo-Aryan	Devanagari	Moderate similarity to verify family-level patterns
Tamil	Dravidian	Tamil	Distant language to test worst-case transfer
Telugu	Dravidian	Telugu	Second Dravidian to verify family patterns

Research Rationale:

- Verify correlation hypothesis: Similar families should show less forgetting
- Test boundary conditions: How dissimilar can languages be and still transfer?

Grammatical Feature Examples

Key Grammatical Cases to Analyze:

1. Word Order:

- Bangla and Hindi: SOV (Subject-Object-Verb)
- Tamil and Telugu: Also SOV
- All four share this feature → May reduce forgetting

2. Postpositions vs Prepositions:

- Bangla: বাড়িতে (barite = house-at) - postposition
- Hindi: घर में (ghar mein = house in) - postposition
- English: in the house - preposition

3. Verb Conjugation:

- Bangla: Complex verb forms with person/tense markers
- Hindi: Similar conjugation patterns
- Tamil: Different conjugation system

Analysis Goal: Measure if models learn these grammatical patterns and whether learning one helps with others

Four Representative NLP Tasks:

1. Sentiment Analysis

- Dataset: IndicNLP sentiment corpus
- Example: Movie/product reviews → Positive/Negative/Neutral

2. Named Entity Recognition

- Dataset: WikiAnn
- Example: Extract Person, Location, Organization from text

3. Natural Language Inference

- Dataset: XNLI
- Example: Does sentence A entail sentence B?

4. News Classification

- Dataset: IndicCorp news articles
- Example: Categorize news into Politics, Sports, etc.

1. MONO (Monolingual Baseline)

- Separate model for each language
- Purpose: Test if after applying continual learning, we can approach MULTI performance

2. MULTI (Multilingual Upper Bound)

- Single model trained on all languages simultaneously
- Maximum positive transfer, no forgetting, Upper bound

3. SEQUENTIAL (Continual Learning Scenarios)

- Train on languages one-by-one: $L1 \rightarrow L2 \rightarrow L3 \rightarrow \text{Bangla}$
- Test multiple orderings (e.g., Hindi $\rightarrow$ Tamil $\rightarrow$ Bangla vs Tamil $\rightarrow$ Hindi $\rightarrow$ Bangla)
- Apply continual learning techniques

Research Design

Compare SEQUENTIAL against MONO and MULTI to measure forgetting and transfer

Continual Learning Technique: Experience Replay

Core Concept: Prevent forgetting by mixing old examples with new ones

How It Works (High-Level):

1. Store a memory buffer of examples from previous languages
2. When training on a new language:
 - 50% of each batch = new language examples
 - 50% of each batch = replayed examples from buffer
3. Model sees old data regularly $\rightarrow$ maintains performance

Why This Method:

- Simple and interpretable (no complex architectures)
- Proven effective in prior work (Winata et al., Rolnick et al.)
- Computationally efficient compared to alternatives
- Easy to implement and reproduce

Future Exploration

If time permits, we may explore learning rate scheduling and curriculum strategies

Transfer Metrics: Measuring Success

1. Forward Transfer (FT): Does prior learning help new languages?

$$FT_i = \text{Accuracy}(\text{after } L_{1:i-1}, \text{test on } L_i) - \text{Accuracy}(\text{random init}, \text{test on } L_i)$$

- Positive FT = Prior languages helped
- Negative FT = Prior languages hurt

2. Backward Transfer (BT): Does new learning hurt old languages?

$$BT_i = \text{Accuracy}(\text{final model}, \text{test on } L_i) - \text{Accuracy}(\text{after } L_i, \text{test on } L_i)$$

- Negative BT = Forgetting occurred
- Positive BT = Later learning improved earlier languages

3. Average Final Performance: Mean accuracy across all languages

4. Per-Task Analysis: Compare metrics across sentiment, NER, NLI, news

Goal: Analyze which language pairs will transfer well

Similarity Metrics:

1. Token Distribution Similarity[2]

- Measure overlap in vocabulary usage
- Higher overlap → more similar languages

2. Lang2vec Syntactic Distance[2]

- Pre-computed linguistic database
- Captures grammar, morphology, word order

3. Language Family

- Indo-Aryan vs Dravidian classification

4. Script Similarity

- Same script family vs different scripts

Execution Timeline: 2-Semester Plan

7th Semester (Months 1-4): Phase 1 and 2

Phase 1: Baseline Establishment (Months 1-2)

- Data collection and preprocessing
- MONO experiments ($5 \text{ languages} \times 4 \text{ tasks} = 20 \text{ models}$)
- MULTI experiments (4 joint models)
- Vanilla sequential baseline (no forgetting mitigation)

Phase 2: Continual Learning Experiments (Months 3-4)

- Implement experience replay system
- Run sequential experiments with multiple orderings
- Compute forward and backward transfer metrics
- Initial analysis of forgetting patterns

8th Semester (Months 5-6): Phase 3

Phase 3: Analysis and Documentation

- Curriculum optimization based on Phase 2 results
- Linguistic feature extraction and correlation analysis
- Thesis writing, visualization, code documentation

Research Exploration: Learning Phase

As part of our learning journey in Natural Language Processing (NLP), we undertook a short research project that culminated in a paper submission to the Bangla Language Processing (BLP) Workshop.

Paper Title: Human-LLM Benchmarks for Bangla Dialect Translation: Sylheti & Chatgaiyya on the BanglaCHQ-Summ Corpus

Project Focus:

- Explored how large language models handle Bangla dialect translation in the medical domain.
- Designed experiments using models such as Qwen, Gemma, GPT-4o, and Gemini.
- Practiced evaluation through BLEU, ROUGE, and METEOR metrics.

Learning Objectives:

- Understand the workflow of conducting an NLP research study—from data preparation to result reporting.
- Gain exposure to prompt design, annotation, and model evaluation.
- Experience the process of writing and formatting an academic paper for submission.

Key Learnings and Reflections

What We Learned:

- Developed a clear understanding of how to structure research methodology and conduct small-scale experiments.
- Learned the importance of reproducibility, evaluation design, and critical analysis in research.
- Improved technical writing, data visualization, and collaborative skills through the paper development process.
- Realized the challenges of working with low-resource languages and the potential of LLMs in addressing them.

Takeaway

This project served as a milestone in our learning and exploration phase, helping us translate theoretical NLP knowledge into practical research experience and preparing us for more advanced future work.

Experimental Framework: Mammoth

Platform: Mammoth Continual Learning Framework

Architecture Overview:

- Backbone: ResNet-18 (vision) / mBERT (NLP)
- Training Loop: Sequential task processing with automatic metric tracking
- Evaluation: Class-IL and Task-IL accuracy after each task

Three Proof-of-Concept Experiments:

1. Catastrophic Forgetting Baseline (Sequential CIFAR-100)
 - Vanilla SGD without replay
2. Continual Learning Validation (CIFAR-100 with DER++)
 - Experience replay with buffer size 500
3. Cross-Lingual Transfer Pilot (Hindi $\rightarrow$ Bangla NER)
 - Sequential NER on WikiANN dataset

Experiment 1: Catastrophic Forgetting Demonstrated

Setup: Sequential CIFAR-100 (10 tasks $\times$ 10 classes each)

Configuration:

- Model: SGD (vanilla fine-tuning, no replay)
- Backbone: ResNet-18
- Learning rate: 0.03, Batch size: 64, Epochs: 5 per task

Results:

Metric	Value	Interpretation
Final Task Accuracy	56.8%	Good on last task
Earlier Tasks	$\approx 0\%$	Complete forgetting
Class-IL (Average)	24.52%	Severe degradation

Key Observation:

- Classifier head completely overwritten by each new task
- Model only predicts classes from the most recent task
- Zero accuracy on tasks 1-9 after training on task 10

Validates that catastrophic forgetting is severe without mitigation

Experiment 2: Continual Learning Effectiveness

Setup: Same Sequential CIFAR-100, now with DER++ (Dark Experience Replay++)

Configuration:

- Model: DER++ with replay buffer (500 examples)
- $\alpha = 0.5$ (distillation weight), $\beta = 0.5$ (replay weight)
- Learning rate: 0.01, Batch size: 64, Epochs: 5 per task

Results Comparison:

Method	Class-IL	Task-IL	Improvement
SGD (Vanilla)	24.52%	24.52%	Baseline
DER++ (Replay)	63.4%	63.4%	+158%

How DER++ Works:

- Replay Buffer: Stores 500 examples with reservoir sampling
- Dual Loss: MSE on saved logits + Cross-entropy on past labels
- Result: Maintains meaningful accuracy on all previous tasks

Proves experience replay effectively mitigates catastrophic forgetting

Experiment 3: Hindi $\rightarrow$ Bangla NER Pilot

Setup: Sequential NER on WikiANN (Hindi $\rightarrow$ Bangla)

Configuration:

- Model: Multilingual BERT (mBERT) with continual learning
- Dataset: WikiANN token sequences (PER, LOC, ORG, O tags)
- Learning rate: 2×10^{-5} , Batch size: 16, Epochs: 5 per language

Results:

Metric	Value	Interpretation
Class-IL (Final)	83.55%	Strong overall performance
Task-IL (Final)	42.1%	Some task confusion
Backward Transfer	+Positive	Bangla improved Hindi!

Key Finding:

- Positive backward transfer: Learning Bangla improved Hindi accuracy
- Suggests shared linguistic features between Indo-Aryan languages
- Validates feasibility of cross-lingual continual learning for Indic languages

Encouraging pilot result for our main thesis experiments

Experiment 4: Cross-Lingual Transfer: Hindi $\rightarrow$ Bangla with IndicBERT

Goal: Use knowledge learned from Hindi to work on Bangla without Bangla training data.

Setup (NLI task):

- Fine-tune `ai4bharat/IndicBERTv2-MLM-only` on Hindi (IndicXNLI-hi)
- Evaluate zero-shot on Bangla (XNLI-bn)
- Labels: entailment, neutral, contradiction

Why transfer works: shared subwords + similar grammar (SOV)

- Bangla: আমি ভাত খেতে চাই (I want to eat rice)
- Hindi: मुझे चावल खाना है (I want to eat rice)

Colab-friendly training:

- `MAX_LEN=256`, `BATCH_SIZE=8`, gradient accumulation, FP16
- 3 epochs, learning rate $2e-5$

Result (zero-shot on Bangla):

- Accuracy $\approx 78.9\%$, Macro-F1 $\approx 78.9\%$

Takeaway

IndicBERT fine-tuned on Hindi generalizes well to Bangla—effective cross-lingual transfer without Bangla training.

The Problem:

- Bangla NLP needs knowledge from multiple languages
- Catastrophic forgetting prevents knowledge accumulation
- No systematic study of forgetting in Indic languages

Our Approach:

- Compare vanilla, cross-lingual, and continual learning strategies
- Systematic experiments with diverse Indic languages
- Analyze linguistic features as transfer predictors
- Optimize language ordering curricula

Expected Impact:

- First comprehensive Indic continual learning benchmark
- Practical guidelines for Bangla NLP development
- Theoretical insights into cross-lingual transfer
- Framework applicable to other language families

Thank You

Questions and Discussion

Contact Information

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